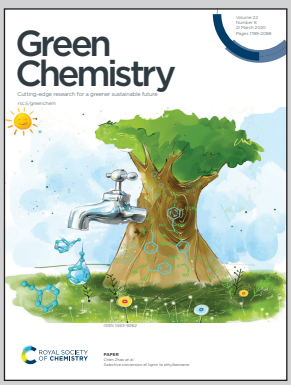


An article presented by Professor Tong-Qi Yuan *et al.* from Beijing Forestry University, China and Professor Run-Cang Sun from Dalian Polytechnic University, China.

In-depth interpretation of the structural changes of lignin and formation of diketones during acidic deep eutectic solvent pretreatment

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In-depth interpretation of the structural changes of lignin and formation of diketones during acidic deep eutectic solvent pretreatment†

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Deep eutectic solvents (DESs) as novel and green solvents can extract high-purity lignin from lignocellulose in a high yield; however, further utilization of the extracted lignin in the DES is a severe challenge in biorefinery processes. Nevertheless, the potential of lignin valorization can be precisely reflected by unveiling the chemical transformations of lignin during DES pretreatment. In this study, we aimed to understand the possible transformation pathway *via* thorough characterization of the regenerated lignin and lignin oil and transformation of lignin into value-added products. During this process, alkali lignin (AL) isolated from poplar was pretreated with choline chloride-lactic acid (ChCl-LA) and choline chloride-oxalic acid (ChCl-OA) at 80–120 °C for 6 hours to produce the regenerated lignin and lignin oil. The yield of the regenerated lignin ranged from 44 to 75% after DES pretreatment. During ChCl-LA pretreatment, γ-acetylated groups in the regenerated lignin were observed at low temperatures and then deacylation occurred at high temperatures, which was beneficial for oxidizing lignin and producing phenolic diketones. Additionally, the depolymerization of lignin fractions was the dominant reaction during DES pretreatment, accompanied by partial condensation reactions. More importantly, this is the first report of obtaining S- and G-derived diketones from lignin oil after DES pretreatment. The degradation mechanism of lignin in the DES system has been proposed. This work opens a way to produce value-added chemicals from lignin in DESs, which needs to be researched in the future.

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Introduction

With the depletion of fossil resources, growing energy demands, and increasing environmental pollution, the utilization of alternative renewable resources has been continuously pursued.[1–3] In view of its availability, renewability, and low cost, lignocellulosic biomass has been recognized as an excellent alternative feedstock.[4] Lignin, one of the three major components of lignocellulose, is the most abundant natural aromatic biopolymer and is widely used to produce valuable aromatic chemicals and functional materials.[5] However, the high-value utilization and efficient conversion of lignin have been severely impeded, because of its complex composition and structural heterogeneity.

Currently, ionic liquids (ILs) have been used to efficiently extract lignin from plant biomass due to their high chemical and thermal stabilities, effective delignification and broad liquid ranges.[6] However, the practical applications of ILs have been hindered due to their incompatibility and high costs.[7] Interestingly, researchers have recently reported that deep eutectic solvents (DESs) show similar physicochemical pro-

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perties to ILs, but DESs are preferred due to their convenient synthesis, wide availability from cheap components, recyclability, and biodegradability.[8,9] More importantly, DESs can selectively separate lignin and cellulose, which is valuable for the successful utilization of the reserved cellulose and regenerated lignin.[10–12] Recently, DESs have been widely and comprehensively studied for biomass refining.[13–16] It has been reported that acidic DESs containing choline chloride (ChCl) as a hydrogen bond acceptor (HBA) have been widely utilized owing to their excellent delignification ability and high efficiency for lignocellulose fractionation among the various DESs reported so far.[17–20] For example, ChCl-lactic acid treatment can extract high-purity lignin (95% from Douglas fir) from woody biomass in a high yield (up to 78% from poplar and 58% from Douglas fir) at 145 °C.[21] In addition, our previous results showed that ChCl-lactic acid treatment performed well at 110 °C for 6 h, and the glucose yield obtained by enzymatic hydrolysis reached 94.3%.[22] Other acidic DES pretreatments, such as ChCl-oxalic acid, are also capable of removing around 70% of lignin.[23] However, lots of research studies have only focused on delignification to enhance the saccharification of the carbohydrate fractions (cellulose and hemicellulose), while the in-depth information about further structural changes and even depolymerization of the regenerated lignin is still underutilized due to its unclearly defined structure and versatility.[24,25] In particular, the catalytic transformation of real lignin by a DES system for producing aromatic molecules has not been reported. Therefore, in-depth structural transformations or even depolymerization of lignin during DES pretreatment can provide guidance for unlocking the potential of using lignin to produce renewable fuels and chemicals. Recent studies have demonstrated that DESs can effectively cleave nearly all β-O-4′ linkages, which greatly facilitates lignin dissolution from lignocellulosic biomass.[26–28] However, only the degradation mechanism of lignin model compounds has been investigated, while the degradation pathways of lignin model compounds and real lignin are usually different in fact, and the interaction mechanism between DESs and lignin has rarely been reported. Due to competitive reactions from residual plant polysaccharides, tracking the chemical transformations of lignin in a DES system is difficult. According to our previous studies, purely alkali lignin from poplar was an appropriate choice for the identification of intermediates and reaction pathways.[29,30] It was relatively free of polysaccharides and also retained its structural integrity.

Based on the above considerations, this work aims to understand in depth the chemical transformations and degradation of the regenerated lignin during DES pretreatment. ChCl-lactic acid (ChCl-LA) and ChCl-oxalic acid (ChCl-OA) (in molar ratios of 1 : 2 and 1 : 1, respectively) were chosen due to their excellent lignin extraction efficiencies. Alkali lignin (AL) isolated from poplar was used to investigate the chemical transformation of lignin during DES pretreatment (80–120 °C, 6 h). The chemical transformations (acylation, condensation, and depolymerization) of lignin fragments during DES pretreatment were systemically investigated by two-dimensional heteronuclear single quantum correlation (2D-HSQC), quantitative ^{13}C-NMR and ^{31}P-NMR, and gel permeation chromatography (GPC). Furthermore, the depolymerized products of lignin oil can be produced and the main aromatic products from lignin were phenolic diketones identified by gas chromatography-mass spectrometry (GC-MS).

Results and discussion

The properties of DESs

To compare the structural changes of DESs before and after pretreatment, the 2D-HSQC spectra were used to analyze their structural features (Fig. S1†). It was revealed that the signals of the pretreated DESs were similar to those of the original DESs, except that the intensities of the signals were slightly different. These results suggest that the chemical structure of the DESs was maintained and stronger hydrogen bonds were formed throughout the DES pretreatment. ^{1}H NMR is another useful tool for confirming the presence of hydrogen bonds between ChCl and a hydrogen bond donor (HBD) (Fig. S2†). It is widely believed that hydrogen bonding leads to the shift of proton signals to higher frequencies (lower field). ^{1}H NMR spectra showed that the protons of the DES in the liquid fraction (rLA-120) experienced upfield shifts (0.2–5.1 ppm) as compared with those of the original DES (ChCl-LA) (0.9–5.3 ppm). This result was probably attributed to the formation of more hydrogen bonds at high temperatures, which was in line with the 2D-HSQC results. Similarly, the protons in the rOA-120 sample shifted to a higher frequency after pretreatment, except that the intensities of the peaks were somewhat weaker, that is, ChCl-OA may be consumed during the pretreatment process. Notably, new signals appeared between 6.0 and 9.0 ppm attributed to the recovered liquids (rLA-120 and rOA-120). These signals are assigned to aromatic protons in S and G units, which confirmed that lignin was degraded during DES pretreatment and the depolymerized products could mainly be S- and G-type aromatics.[31,32] Therefore, it is necessary to analyze the structure of the regenerated lignin to further verify the decomposition products obtained during DES pretreatment.

Structural characterization of the regenerated lignin

Based on the survey results, alkali lignin (AL) isolated from poplar was treated with acidic DESs at three different temperatures (80, 100, and 120 °C) for 6 h to investigate the depolymerization mechanism of lignin. Fig. 1 shows that the yield of the regenerated lignin fractions obviously decreased after DES pretreatment, suggesting that DESs efficiently degraded a portion of the lignin fragments. Meanwhile, the sample yield gradually decreased with an increase in temperature, especially at 120 °C. In addition, ChCl-OA pretreatment resulted in a lower yield of the regenerated lignin at the same temperature compared to ChCl-LA pretreatment. The Kamlet–Taft solvatochromic method is an informative tool for assessing the chemical properties of DESs. To better understand the effect of DESs on

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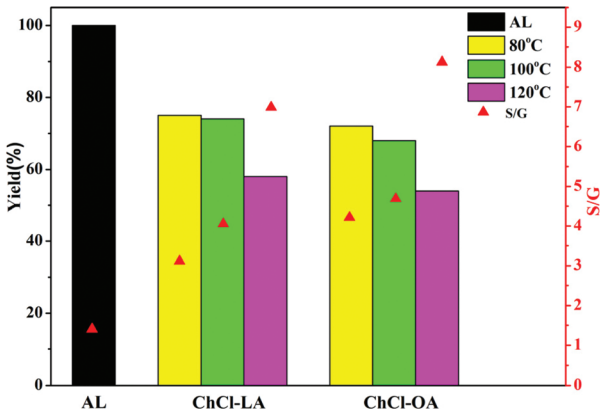


Fig. 1 Yield of AL and the regenerated lignin fractions during DES pretreatment.

the degradation of lignin, the Kamlet–Taft empirical parameters (acidity α, basicity β, and polarizability π^*) were determined (Table S1†). After analyzing the data, we speculated that the lower yield of the ChCl-OA lignin possibly originated from the higher α–β value of ChCl-OA (1.10) than that of ChCl-LA (0.80). Mu *et al.*[9] also found that the solubility of lignin was positively correlated with the α–β value of ChCl-LA. These results indicated that DESs with a high α–β value can easily form strong hydrogen bonds with lignin, thus facilitating the degradation of lignin.

2D-HSQC NMR can determine the specific carbon–hydrogen functionalities to identify the structural variations in lignin, which cannot be determined by ^{13}C and ^{1}H NMR. The side-chain (δ_C/δ_H 50–90/2.5–6.0) and aromatic (δ_C/δ_H 100–135/6.0–8.0) regions in the 2D-HSQC NMR spectra of lignin fractions before and after DES pretreatment are shown in Fig. 2 and 3. The peak assignments are listed in Table S2,† according to the previous papers.[33–35] After ChCl-LA pretreatment, the peaks assigned to γ-acetylated β-O-4′ linkages (A'_γ, 63.0/4.36 ppm) were observed at 80 and 100 °C, implying that an acylation reaction occurred between the γ-OH group in the lignin substructure and the COOH groups in the LA. This phenomenon was in line with the previous literature, which also showed the presence of an A'_γ signal after ChCl-LA pretreatment.[22,28,36] Nevertheless, it was worth noting that the A'_γ signal completely disappeared when the temperature reached 120 °C, which has been rarely observed during the ChCl-LA pretreatment process. This finding indicated that deacylation occurred at high temperatures, likely because of the lack of interference from polysaccharides. This could be beneficial for oxidizing lignin and producing phenolic diketones. However, after ChCl-OA pretreatment, the γ-acetylated β-O-4′ groups were not detected in the regenerated lignin samples, possibly due to the higher acid strength of OA (pK_{a1} = 1.25 and pK_{a2} = 4.14) than LA (pK_a = 3.86) or the higher α–β value of ChCl-OA.

In addition to acylated lignin structures, the condensation of some lignin fractions could also take place after DES pretreatment. It has been reported that condensation reactions occur not only at the 2-, 5-, and 6-positions of G units, but also between the 5-position of other aromatic rings and C_α in lignin side chains.[30,37] As shown in Fig. 3, the chemical shifts

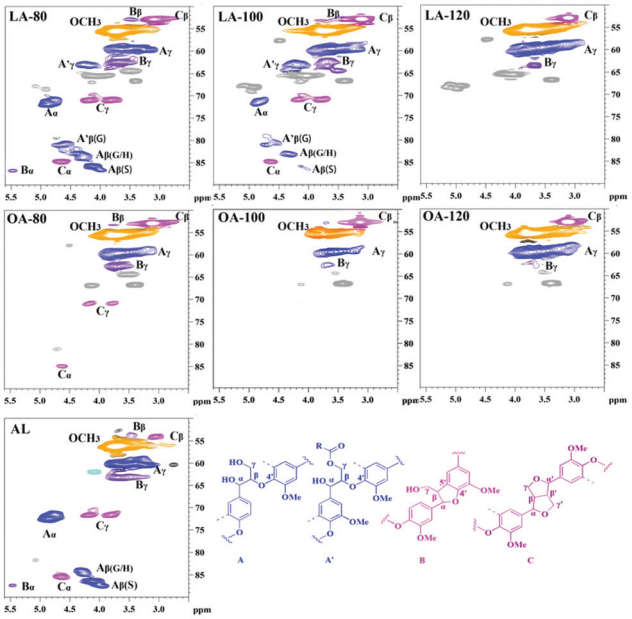


Fig. 2 Side-chain regions in the 2D-HSQC NMR spectra of AL and the regenerated lignin fractions.

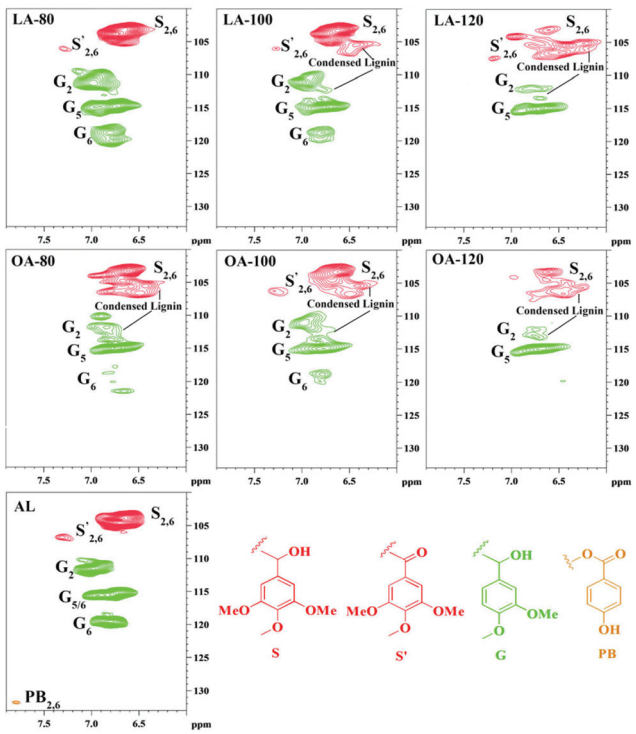


Fig. 3 Aromatic regions in the 2D-HSQC NMR spectra of AL and the regenerated lignin fractions.

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of G units (especially G_2 and G_6) were altered during DES pretreatment due to the formation of intricate C–C linkages during acid-catalyzed condensation.[38] Meanwhile, the signals corresponding to the original S- and G-type lignin structures gradually decreased, while some related signals from the condensed S and G units appeared after DES pretreatment. Moreover, variations in the S/G ratio were helpful for evaluating the structural transformation of lignin (Fig. 1). AL was determined to have a ratio of S/G subunits of 1.41, while the S/G ratio sharply increased as a function of temperature after DES pretreatment. The increased S/G ratio indicated that G-type lignin units were preferentially removed or degraded in the prepared DES (discussed below). Besides, the S/G ratios of the regenerated lignin after ChCl-OA pretreatment were higher than those after ChCl-LA pretreatment at the same temperature, suggesting that the higher α–β value of ChCl-OA promoted the degradation of G-type lignin units. In addition, the occurrence of condensation reactions of G units could also lead to an increase in the S/G ratio. ^{13}C NMR spectroscopy is frequently used to quantify the side chain linkages and different condensed moieties (Fig. 4). A detailed comparison of the integration values in the aromatic regions, including the protonated (C–H), condensed (C–C), and oxygenated (C–O) regions, can provide additional evidence for the condensation of lignin fractions. After DES pretreatment, the amount of C–C aromatics remarkably increased (1.95/Ar in LA-120 and 2.20/Ar in OA-120), while the amount of C–H aromatics decreased (2.11/Ar in LA-120 and 2.17/Ar in OA-120). These results demonstrated that DES pretreatment resulted in the condensation of lignin fractions, which agreed with the 2D-HSQC analysis. Notably, the distinct signals between 169.9 and 176.2 ppm and at 20.2 ppm in the LA-120 spectrum are assigned to C=O in the carboxyl group, implying that the oxidation reactions may occur after ChCl-LA pretreatment. The 2D-HSQC spectra revealed that the intensity of the A_α signal gradually decreased and even disappeared after DES pretreatment, which probably could be attributed to the oxidation of C_α OH groups. In addition, the CH_3O content decreased sharply from 1.65/Ar in AL to 1.39/Ar in OA-120 and 1.49/Ar in LA-120, indicating that demethoxylation also took place during the pretreatment.

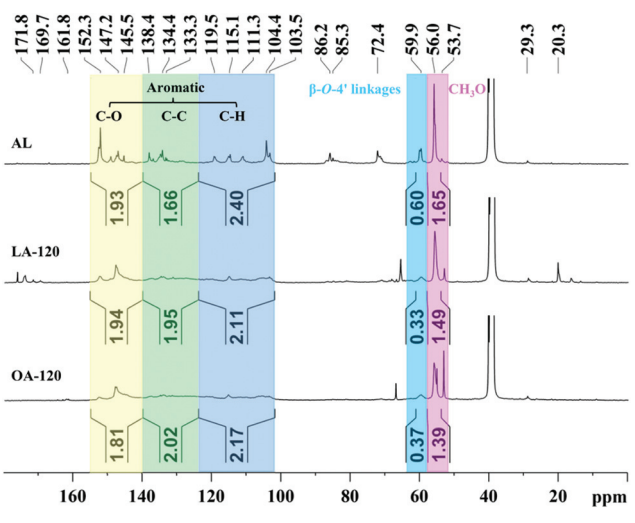


Fig. 4 ^{13}C-NMR spectra of AL and the regenerated lignin fractions at 120 °C (results expressed per Ar).

Except for acylation and condensation, depolymerization reactions were also observed during DES pretreatment, particularly the cleavage of ether bonds. The content of hydroxyl groups in native lignin is low, but it can be increased by the cleavage of ether bonds and other depolymerization reactions. This suggests that changes in the hydroxyl groups of lignin can indirectly demonstrate structural variations in lignin. ^{31}P NMR spectroscopy is a convenient tool to quantify different functional groups in the lignin fractions, especially hydroxyl groups (Fig. S4† and Table 1). It was observed that the content of total phenolic OH in ChCl-LA lignin samples sharply increased with an increase in temperature and reached 3.69 mmol g^{-1} in LA-120, which was nearly three times higher than that of AL (1.18). The increased phenolic OH was mainly ascribed to the cleavage of β-O-4′ bonds after DES pretreatment. The content of total phenolic OH in ChCl-OA lignin samples increased compared with that of AL, while it increased at 80 °C and then decreased at 100 and 120 °C. This phenomenon suggested that depolymerisation and dehydration or condensation occurred simultaneously at high temperatures. As revealed by 2D-HSQC NMR spectroscopy, the content of β-O-4′ bonds considerably decreased with an increase in temperature (Table S3†), and the signals corresponding to A_β(G/H) (δC/δH 83.9/4.29) and A_β(S) (δC/δH 85.9/4.11) finally disappeared (Fig. 2). Moreover, the signal intensity of β-O-4′ linkages and C–O aromatics clearly decreased in the regenerated lignin, as is also shown in the ^{13}C-NMR spectra. These results proved that DES pretreatment promoted the depolymerization of lignin fractions, especially the cleavage of β-O-4′ linkages. In addition, the aliphatic OH content remarkably decreased compared with that of AL (3.81 mmol g^{-1}), due to rapid dehydration during the cleavage of β-O-4′ linkages and condensation. It is worth noting that the content of carboxylic groups in the ChCl-LA lignin was much higher than that of AL. It was reasonable to deduce that the increase in carboxylic groups was caused by a small amount of residual LA because a minor LA correlation was detected in the ChCl-LA lignin by 2D-HSQC analysis (Fig. S3†). Another reason could be that aliphatic OH groups were oxidized into carboxylic groups during the pretreatment. However, the content of carboxylic groups in the ChCl-OA lignin was nearly unchanged compared with that in AL, likely attributed to the strong steric hindrance of OA.

Different molecular weights of lignin can give important insights into the dissociation and condensation of lignin after DES pretreatment. The AL and regenerated lignin samples were acetylated and then analyzed by GPC (Table 2). The M_w (number average molecular weight) of AL was 5080 g mol^{-1} and M_w/M_n (polydispersity index, PDI) was 1.59. The aforementioned NMR analysis suggested that the depolymerization and condensation of lignin were two competing reactions during

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Table 1 Quantification of the functional groups in the lignin samples by quantitative ^{31}P-NMR (mmol g^{-1})

Samples	Carboxylic group	Aliphatic OH	Syringyl OH	Guaiacyl OH		*p*-Hydroxyphenyl OH	Total phenolic OH
				C[a]	NC[b]		
AL	0.22	3.81	0.42	0.11	0.50	0.15	1.18
LA-80	0.65	4.40	0.50	0.11	0.66	0.07	1.34
LA-100	1.68	4.55	1.20	0.20	0.67	0.10	2.17
LA-120	1.06	2.20	2.15	0.45	0.87	0.22	3.69
OA-80	0.27	1.79	2.04	0.34	1.04	0.17	3.59
OA-100	0.20	0.87	1.24	0.25	0.54	0.11	2.14
OA-120	0.21	0.77	1.50	0.34	0.78	0.28	2.90

[a] C, condensed. [b] NC, non-condensed.

Table 2 Weight-average (M_w) and number-average (M_n) molecular weights (g mol^{-1}) and polydispersity (M_w/M_n) of lignin

Samples	AL	ChCl-LA			ChCl-OA		
		80 °C	100 °C	120 °C	80 °C	100 °C	120 °C
M_w	5080	4420	1790	1830	1300	1300	1800
M_n	3190	2380	1430	1470	920	920	1410
M_w/M_n	1.59	1.56	1.25	1.24	1.42	1.42	1.27

the pretreatment with DESs. However, it was found that the M_w of lignin samples dramatically decreased (1300–1800 g mol^{-1}) at high temperatures. The lower M_w could be caused by the low content of β-O-4′ bonds (determined by ^{31}C NMR and 2D-HSQC) and the high content of phenolic OH (quantified by ^{31}P NMR). These results suggested that the depolymerization of lignin fractions was the dominant reaction under acidic conditions, accompanied by partial condensation reactions. Moreover, the PDI of the regenerated lignin was lower (1.24–1.56) compared to that of AL (1.59), indicating that DES pretreatment facilitated homogeneity in the molecular weight of lignin. In addition, the thermal stability of the regenerated lignin was significantly enhanced after DES pretreatment, because the decomposition temperature of the regenerated lignin shifted to a higher value, and its residual mass slightly decreased (Fig. S6†).

Characterization of lignin oil

After the precipitation of large amounts of lignin fractions as regenerated lignin, small amounts of depolymerized lignin fragments remained in the solution. To further understand the inherent mechanism of lignin depolymerization, the depolymerization products of lignin after DES pretreatment were identified by GC-MS (Table S4 and Fig. S7†). Due to no standard mass spectrometry of S- and G-derived diketones, we confirmed these two products by comparing with the standard mass spectrum in previous reports;[39,40] other lignin monomer products were identified using the NIST 2011 spectral library. As listed in Table S4,† the identified depolymerized products were primarily composed of S- and G-derived phenolics containing ketones and aldehydes. Among these phenolic monomers, S- and G-derived diketones were two major depolymerization products (Fig. 5), which have not been previously reported in lignin pretreatment in a DES system. This is the first time that a DES system has been utilized to obtain certain diketone compounds. The resulting diketone compounds have various applications, including as functional materials in synthetic phenolic resins, versatile intermediates, synthons in synthetic chemistry, antibiotics, and ligands.[41–43] However, it should be noted that the peak intensity of some phenolic compounds was relatively low overall, implying that the yield of the obtained aromatic products from the degradation of lignin in

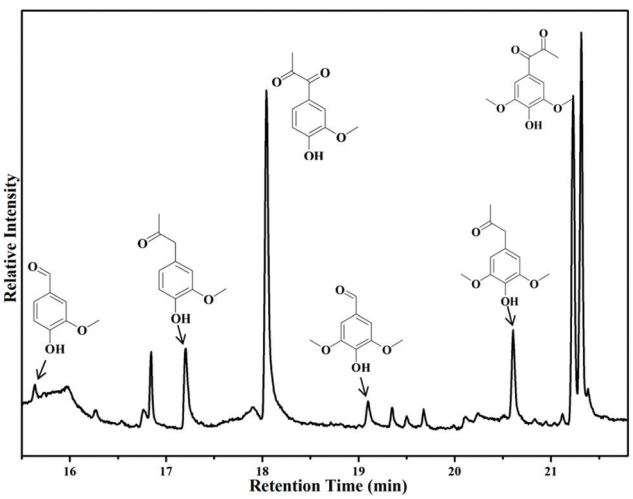


Fig. 5 Gas chromatogram of lignin oil obtained after ChCl-OA pretreatment at 100 °C.

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the DES system can be further improved. In addition, the GC-MS results suggested that the pretreatment temperature had an enormous effect on the depolymerized products (Fig. S7†). It was observed that few depolymerized products were detected at a mild temperature during pretreatment, such as 80 °C. When the temperature increased, the peak intensity and types of lignin monomeric products increased, indicating that higher temperatures facilitated the depolymerization of lignin, and aromatic compounds were simultaneously formed.

Mechanism studies of lignin

To achieve the high value potential of lignin, it is important to explore the transformation mechanism of lignin in the DES system. Combining all the data herein, it is confirmed that the DES pretreatment led to the cleavage of ether bonds *via* two feasible routes (Fig. S9†). Route A involves the removal of the C_α alcohol (indirectly quantified by ^{31}P NMR) and the formation of highly reactive benzylic carbocations in the lignin side chains. FTIR analyses show that a strong peak at 1738 cm^{-1} (the unconjugated carbonyl and ester group) appeared and a peak from AL at 1657 cm^{-1} (the conjugated carbonyl group) disappeared (Fig. S5†), indicating the damage of the C_α double bond and the destruction of the conjugated structure.[28] Such a pathway is the most prevalent mechanism of lignin depolymerization during acidic pretreatment.[44,45] Route B leads to the oxidation of the C_α position and the acylation of the C_γ position (as shown by 2D-HSQC NMR). Attenuated total reflectance Fourier transform infrared (ATR-FTIR) spectra were regularly measured to detect chemical structural changes of liquid fractions at different times. These results showed that the signal of the conjugated carbonyl stretch (1655 cm^{-1}) gradually appeared over time during DES pretreatment, confirming that OH groups at the C_α position were oxidized. Stahl *et al.*[46] demonstrated that the C_α ketone was crucial to promote the cleavage of β-O-4′ linkages and the mechanism for C–O cleavage involved in formylation, elimination and hydrolysis. Aside from the cleavage of β-O-4′ bonds, other possible reactions, such as dehydration, acylation, demethoxylation, and condensation, are also demonstrated during DES pretreatment, as shown in Fig. 6.

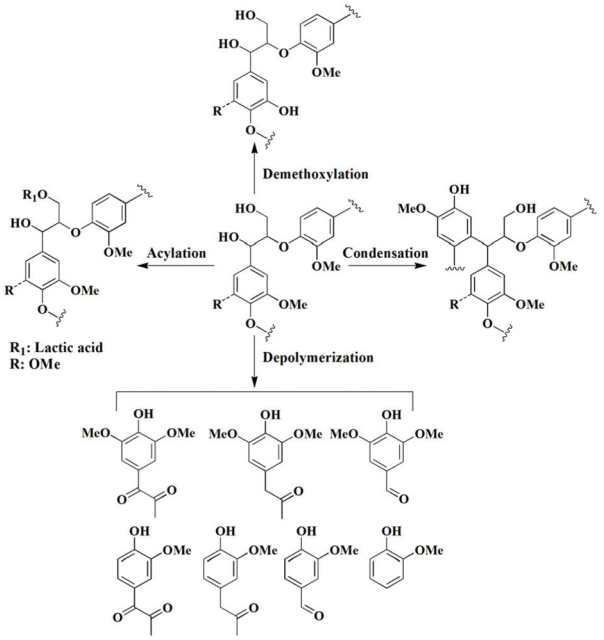


Fig. 6 Mechanism studies of lignin during DES pretreatment.

Experimental

The poplar (triploid of *Populus tomentosa* Carr.) was ground to pass through a 0.8 mm size screen and was extracted with toluene–ethanol (2 : 1 v/v) in a Soxhlet extractor for 6 h. The extracted poplar sawdust was milled for 1 h in a 500 mL ZrO_2 bowl with mixed balls. The ball-milled poplar was treated with 1% sodium hydroxide with a solid-to-liquid ratio of 1 : 15 (g mL^{-1}) at 75 °C for 3 h. According to our previous study,[30] the composition of the alkali lignin is lignin (97.3%), total carbohydrates (1.1%), and others (1.6%).

0.3 g of AL was transferred into a pressure tube. Then, the DES (6.0 g) was slowly added into the tube. The mixture was then placed in an oil bath and heated on a hot plate (IKA RCT basic, Germany) with vigorous magnetic stirring (400 rpm) under different conditions (80–120 °C, 6 h). After the pretreatment, the regeneration of the dissolved lignin was carried out by precipitation of DES–lignin solution into 120 mL deionized water under vigorous magnetic stirring. The regenerated lignin was separated by centrifugation and washed with acidic water (pH = 2.0) three times and freeze-dried. The regenerated lignin was labelled as XX-YY (*e.g.* LA-80). The recovered liquid was labelled as rXX-YY (*e.g.* rLA-80). The recovered liquid fraction was extracted using ethyl acetate. Ethyl acetate was removed by reduced pressure distillation and lignin oil was obtained. Lignin oil was labelled as Z-YY (*e.g.* L-80). XX and Z refer to the DES type and YY refers to the pretreatment temperature. The regenerated lignin was characterized by two-dimensional heteronuclear single quantum correlation (2D-HSQC), quantitative ^{13}C-NMR and ^{31}P-NMR, and gel permeation chromatography (GPC). Lignin oil was accurately evaluated by gas chromatography-mass spectrometry (GC-MS) and attenuated total reflectance Fourier transform infrared (ATR-FTIR) spectroscopy. The detailed procedures are listed in the ESI.†

Conclusions

This study confirmed the effect of DESs on the structural transformations of lignin and provided the first report of obtaining phenolic diketones from lignin oil after DES pretreatment. During ChCl-LA pretreatment, γ-acetylated groups in the regenerated lignin were observed at low temperatures, while deacylation occurred at high temperatures. Partial condensation simultaneously occurred, which improved the thermal stability of the regenerated lignin. Moreover, DES pre-

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treatment promoted the cleavage of large amounts of ether bonds (especially β-O-4′ linkages), leading to an increase in the number of phenolic OH groups and lower molecular weights. More importantly, phenolic diketones in lignin oil after DES pretreatment were obtained for the first time. Based on the comprehensive analysis, the mechanism of the cleavage of β-O-4′ linkages during DES pretreatment could have two feasible routes: the removal of C_α OH groups to form benzylic carbocations and the oxidation of OH groups to obtain C_α ketones. This study provides a deeper insight into the chemical transformations of lignin in the DES system, which can open new perspectives for delignification and lignin conversion processes.

Conflicts of interest

There is no conflict to declare.

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